

B.2.4 Exam Practice 4: Virtualization and Cloud Computing

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Score: 80%

Passing Score: 80%



Question 1

✕ Incorrect

What does the Microsoft Windows 'Programs and Features' applet allow the user to manage?

☒ Security requirements ✕ Incorrect

☐ Test development

☐ Resource requirements

☐ Legacy software/OS ✓ Correct

Explanation

Legacy software applications can be managed through the Microsoft Windows 'Programs and Features' applet.

Resource requirements are important to consider and meet when setting up a virtualized system and network for virtualized systems to use. If the system is not designed with resource requirements in mind, it will be overwhelmed and dysfunctional.

Security requirements are incorrect because the amount of central processing units (CPUs) and random access memory (RAM) a system needs is not going to impact the security of a system.

Test development often uses virtual desktop platforms based on some sort of guest OS hypervisor.

Related Content



8.1.2 Uses for Virtualization

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Question 2

✔ Correct

A cloud service provider (CSP) works with several organizations to establish new cloud deployments that best match their needs. Each organization has similar security needs and is open to sharing resources with other organizations to save money. Which cloud deployment sounds best for the CSP's new customers?

- ☐ Hybrid cloud
- ☐ Public cloud
- ☒ Community cloud ✔ Correct
- ☐ Private cloud

Explanation

The community will be the best deployment model for the CSP's new customers. A community cloud allows several organizations to share the costs of either a hosted private or fully private cloud by pooling resources for common concerns, such as standardization and security policies.

Private cloud is incorrect because it does not allow for sharing cloud resources with organizations.

Hybrid cloud deployment is not the best choice for the CSP's new customers because the customers want to share resources full-time.

The public cloud is incorrect because the CSP's customers still want a private cloud to share with organizations with the same security concerns.

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8.2.3 Common Cloud Deployment Models

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Question 3

✔ Correct

A large store chain is eager to reduce the cost of maintaining its servers but is unsure what the best method would be for its use case. The major requirements for the chosen solution are that the store chain needs to own its cloud infrastructure (without sharing with another organization) to maintain maximum privacy and security of the data. Which cloud computing option best meets the store chain's major requirements?

☒ Private cloud ✔ Correct

☐ Hybrid cloud

☐ Public cloud

☐ Community cloud

Explanation

The private cloud meets the store chain's two major requirements because it gives organizations greater control over the privacy and security of its data stored in the cloud.

The public cloud is incorrect because there are risks regarding performance and security. The public cloud is a shared resource with other organizations, and a cloud service provider (CSP) runs it.

A hybrid cloud is incorrect because some cloud functions can be done via a public cloud, creating potential vulnerabilities to privacy and security.

Community cloud is incorrect because it requires the store chain to share a hosted private or fully private cloud with other organizations.

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8.2.3 Common Cloud Deployment Models

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Question 4

✔ Correct

An on-premises data center is moving to a cloud model to take advantage of the many benefits cloud computing offers. One of the most important benefits of this specific data center is handling changes to demand in real-time without degrading services. Which of the following is the most important benefit of migrating this data center to the cloud?

☒ Rapid elasticity ✓ Correct

☐ High availability

☐ Shared resources

☐ Metered utilization

Explanation

Rapid elasticity refers to the system's ability to handle changes to demand in real-time without loss of service or performance. Rapid elasticity allows a system to reduce costs when demand is low.

Metered utilization is the per-use billing for resources consumed by the cloud, making cloud offerings cheaper or more expensive for an organization, depending on their resource consumption.

Shared resources are incorrect. To meet availability, scalability, and elasticity requirements, cloud providers must be able to provision and de-provision resources automatically through the pooling of shared resources and virtualization.

High availability means that the service experiences very little downtime. Downtime can occur as a result of scheduled maintenance and unexpected outages.

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8.2.1 Cloud Characteristics

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Question 5

✔ Correct

While planning a new virtual environment, a systems engineer noted that one virtual machine (VM) would run a Windows OS and store two snapshots at all times. The VM will need to have at least 80 GB in its hard disk drive for the VM to function properly. What is the systems engineer considering when planning this aspect of the new virtual environment?

☒ Resource requirements ✓ Correct

☐ Security requirements

☐ Legacy software/OS

☐ Test development

Explanation

Resource requirements are important to consider and meet when setting up a virtualized system and network for virtualized systems to use. If the system is not designed with resource requirements in mind, it will be overwhelmed and dysfunctional.

Security requirements are incorrect because the amount of central processing units (CPUs) and random access memory (RAM) a system needs will not impact the security of a system.

Legacy software applications and operating systems may not perform well if a user's system has been upgraded.

Test development often uses virtual desktop platforms based on some sort of guest OS hypervisor.

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Question 6

✔ Correct

A holiday decor store typically runs its cloud infrastructure privately, but its servers get bogged down by transactions during the peak holiday season. To mitigate any revenue losses, the store must outsource processing power to an outside cloud service provider to help support the increased sales volume. What cloud deployment model does the holiday decor store utilize?

- ☐ Public cloud
- ☐ Private cloud
- ☐ Community cloud
- ☒ Hybrid cloud ✓ Correct

Explanation

Hybrid cloud deployment is what the holiday decor store utilizes because it allows the store to utilize its private cloud infrastructure during non-peak sales times and then incorporate a public cloud for support during peak sales times.

Private cloud is incorrect because it does not allow the decor store to outsource resources during peak sales.

The public cloud is incorrect because it does not allow the decor store the opportunity to use its internal cloud infrastructure.

Community is not the best answer because the decor store does not want to share resources with other organizations year-round.

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8.2.3 Common Cloud Deployment Models

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Question 7

✕ Incorrect

What features of the cloud allow multiple users to simultaneously access content and work collaboratively while tracking edits by each user?

☒ Shared resources ✕ Incorrect

☐ High availability

☐ File synchronization ✓ Correct

☐ Community cloud

Explanation

File synchronization is an advantage of cloud storage that replicates files between data centers and different devices such as PCs and smartphones. Cloud storage also allows multiple users to access content and work collaboratively simultaneously.

To meet availability, scalability, and elasticity requirements, cloud providers must be able to provision and de-provision resources automatically through the pooling of shared resources and virtualization.

A community cloud is a cloud environment deployed for shared use by cooperating tenants. These tenants usually have similar security and operating concerns.

High availability means that the service experiences very little downtime. Downtime can occur because of scheduled maintenance and unexpected outages.

Related Content

-  2.2.4 Motherboard Storage Connectors
-  3.2.3 Solid-State Drives
-  3.2.11 Optical Drives
-  4.2.13 Troubleshooting Problems with Storage Drives & RAID Arrays
-  8.2.7 Cloud File Storage
-  12.4.2 Storage
-  15.2.2 Windows Domains and Active Directory
-  21.1.4 Backup Media Requirements

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Question 8

✕ Incorrect

What is the typical use for virtual desktop platforms based on guest OS hypervisor?

☒ Cross-platform virtualization ✕ Incorrect

☐ Legacy software/OS

☐ Testing and development ✓ Correct

☐ Resource requirements

Explanation

Desktop virtual platforms, usually based on guest OS, are typically used for testing and development.

Resource requirements are important to consider and meet when setting up a virtualized system and network for virtualized systems to use. If the system is not designed with resource requirements in mind, it will be overwhelmed and dysfunctional.

Cross-platform utilization tests software applications under different OSs and resource constraints and is important to ensure the application is compatible across several different environments.

Legacy software applications and operating systems may not perform well if a user's system has been upgraded.

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8.1.2 Uses for Virtualization

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Question 9

✔ Correct

Sandboxes can be used to analyze viruses, worms, and Trojans. Sandboxes can also provide an isolated environment for what other important function?

- ☐ Cross-platform virtualization
- ☒ Testing and development ✓ Correct
- ☐ Legacy software/OS
- ☐ Resource requirements

Explanation

Test development often uses virtual desktop platforms based on some guest OS hypervisor.

Resource requirements are important to consider and meet when setting up a virtualized system and network for virtualized systems to use. If the system is not designed with resource requirements in mind, it will be overwhelmed and dysfunctional.

Cross-platform utilization tests software applications under different OSs and resource constraints and is important to ensure the application is compatible across several different environments.

Legacy software applications and operating systems may not perform well if a user's system has been upgraded.

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Question 10

✔ Correct

When a user downloads a Microsoft Store application, where does the application run to prevent the app from making system-wide changes and prevent a faulty store app from "crashing" the entire OS?

- ☐ Test development
- ☐ Resource requirements
- ☒ Sandbox ✓ Correct
- ☐ Cross-platform virtualization

Explanation

A sandbox is an isolated environment where Microsoft Store apps run to prevent the app from making system-wide changes and prevent a faulty store app from "crashing" the entire OS.

Test development often uses virtual desktop platforms based on some sort of guest OS hypervisor.

Cross-platform utilization tests software applications under different OSs and resource constraints and is important to ensure the application is compatible across several different environments.

Resource requirements are important to consider and meet when setting up a virtualized system and network for virtualized systems to use. If the system is not designed with resource requirements in mind, it will be overwhelmed and dysfunctional.

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Question 11

✔ Correct

What software delivery method enables programmers and application administrators to ensure that the application used by clients is always updated with the latest code?

- ☐ Legacy software/OS
- ☐ Sandbox
- ☐ Cross-platform virtualization
- ☒ Application virtualization ✓ Correct

Explanation

Application virtualization is a software delivery method that enables programmers and application administrators to ensure that clients' applications are always updated with the latest code.

Cross-platform utilization tests software applications under different OSs and resource constraints and is important to ensure the application is compatible across several different environments.

A sandbox is an isolated environment to analyze viruses, worms, and Trojans in a space that prevents the researcher's computer and network from becoming infected.

Legacy software applications and operating systems may not perform well if a user's system has been upgraded.

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8.1.2 Uses for Virtualization

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Question 12

✔ Correct

An important part of testing a new application is running it under several different OSs and resource constraints so the developer can be sure it will work for all customers regardless of their system. What term is used to identify the application testing described in this question?

- ☐ Resource requirements
- ☐ Test development
- ☒ Cross-platform virtualization ✔ Correct
- ☐ Sandbox

Explanation

Cross-platform virtualization tests software applications under different OSs and resource constraints and is important to ensure the application is compatible across several different environments.

A sandbox is an isolated environment to analyze viruses, worms, and Trojans in a space that prevents the researcher's computer and network from becoming infected.

Test development often uses virtual desktop platforms based on some guest OS hypervisor.

Resource requirements are important to consider and meet when setting up a virtualized system and network for virtualized systems to use. If the system is not designed with resource requirements in mind, it will be overwhelmed and dysfunctional.

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Question 13

✔ Correct

Cloud service providers (CSPs) help organizations save time and money by hosting virtual resources on their hardware. What do CSPs use to ensure efficiency while providing services to customers?

- ☐ Community cloud
- ☐ Public cloud
- ☒ Shared resources ✔ Correct
- ☐ Hybrid cloud

Explanation

Shared resources are correct. To meet availability, scalability, and elasticity requirements, cloud providers must be able to provision and de-provision resources automatically through the pooling of shared resources and virtualization.

Community cloud is incorrect. CSPs provide community clouds to customers using shared resources. A community cloud is a way for organizations to share a private cloud and save money.

The public cloud is incorrect. CSPs provide public clouds to customers using shared resources. CSP offers public cloud on a subscription or pay-as-you-go basis but comes with security and performance risks.

Hybrid cloud is incorrect. CSPs provide hybrid clouds to customers using shared resources. Customers using hybrid clouds can use public cloud resources when their private infrastructure is insufficient.

Related Content[8.2.3 Common Cloud Deployment Models](#)[resources\questions\pt_core1_220_1201_4_2_009.question.xml](#)

Question 14

✔ Correct

Remote wipe, locator apps, device encryption, remote backup, and antivirus software fall under which category of requirements for virtualized environments?

- ☐ Test development
- ☐ Legacy software/OS
- ☐ Resource requirements
- ☒ Security requirements ✓ Correct

Explanation

Security requirements are correct because all the things listed in the question refer to profile security requirements established for any system.

Resource requirements are important to consider and meet when setting up a virtualized system and network for virtualized systems to use. If the system is not designed with resource requirements in mind, it will be overwhelmed and dysfunctional.

Test development often uses virtual desktop platforms that are based on some guest OS hypervisor.

Legacy software applications and operating systems may not perform well if a user's system has been upgraded.

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8.1.5 Virtualization Security Requirements

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Question 15

✔ Correct

Cloud provisioning models have unique characteristics that make them better or worse than on-premises models, depending on an organization's needs. Which characteristic refers to the per-use billing for resources consumed by the cloud?

- ☐ Shared resources
- ☒ Metered utilization ✓ Correct
- ☐ Rapid elasticity
- ☐ Community cloud

Explanation

Metered utilization is the per-use billing for resources consumed by the cloud, making cloud offerings cheaper or more expensive for an organization, depending on their resource consumption.

Shared resources are incorrect and can be thought of as the resources measured in this question.

Rapid elasticity is incorrect. Rapid elasticity refers to the system's ability to handle changes and demand in real-time without degrading services. Conversely, rapid elasticity can result in cost savings by reducing resources when demand is low.

Community cloud is incorrect. A customer using a community cloud could be subject to metered utilization of the cloud resources.

Related Content[8.2.1 Cloud Characteristics](#)[resources\questions\pt_core1_220_1201_4_2_010.question.xml](#)

Question 16

✔ Correct

A startup company does not have the funds or the time to buy and configure its computer resources, so it pays a service provider to deploy IT resources such as servers, load balancers, and storage components. What service is the startup company paying for?

- ☐ Platform as a service (PaaS)
- ☐ Software as a service (SaaS)
- ☒ Infrastructure as a service (IaaS) ✔ Correct
- ☐ Shared resources

Explanation

Infrastructure as a service (IaaS) is correct. IaaS is a means of provisioning IT hardware and the internet links they require, as needed, from a service provider's data center.

Software as a service (SaaS) is incorrect. SaaS is a means of purchasing software licenses for a given number of users and does not include IT hardware resources.

Platform as a service (PaaS) is incorrect. PaaS is incorrect because the question does not mention that the company needs IT hardware resources and multi-tier web applications and database platforms on top of the provided IT hardware resources.

Shared resources are not a service that can be purchased. Service providers can provide services using shared resources to lower the cost to the customer.

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8.2.5 Common Cloud Service Models

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Question 17

✔ Correct

A systems engineer is building a virtual machine to run a service with an 8 core CPU and 500 GB of random access memory to function properly. Where would these two hardware requirements be listed?

- ☐ Legacy software/OS
- ☐ Test development
- ☒ Resource requirements ✓ Correct
- ☐ Security requirements

Explanation

Resource requirements are important to consider and meet when setting up a virtualized system and network for virtualized systems to use. If the system is not designed with resource requirements in mind, it will be overwhelmed and dysfunctional.

Security requirements are incorrect because the amount of central processing units (CPUs) and random access memory (RAM) a system needs will not impact the security of a system.

Legacy software/OS is incorrect because a list of legacy software and OSs would not encompass hardware like CPU and RAM.

Test development often uses virtual desktop platforms based on some guest OS hypervisor.

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Question 18

✔ Correct

What can a cloud developer use to analyze viruses, worms, and Trojans without infecting their computer and network?

- ☒ Sandbox ✓ Correct
- ☐ Test development
- ☐ Resource requirements
- ☐ Cross-platform virtualization

Explanation

A sandbox is an isolated environment to analyze viruses, worms, and Trojans in a space that prevents the researcher's computer and network from becoming infected.

Test development often uses virtual desktop platforms based on some guest OS hypervisor.

Cross-platform utilization tests software applications under different OSs and resource constraints and is important to ensure the application is compatible across several different environments.

Resource requirements are important to consider and meet when setting up a virtualized system and network for virtualized systems to use. If the system is not designed with resource requirements in mind, it will be overwhelmed and dysfunctional.

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8.1.2 Uses for Virtualization

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Question 19

✔ Correct

A software development company uses a system to allow developers to log into any computer throughout the building and load their unique desktop environment. The IT department decided to manage these servers in the data center located in the basement of the company's office building. What system is the software development company leveraging?

- ☐ Infrastructure as a service (IaaS)
- ☒ Virtual desktop infrastructure (VDI) on premises ✓ Correct
- ☐ Virtual desktop infrastructure (VDI) in the cloud
- ☐ Platform as a service (PaaS)

Explanation

Virtual desktop infrastructure (VDI) on-premises is correct because the developers log into virtual machines that load the user's computer images. The servers are hosted on the same network as the clients.

VDI in the cloud is incorrect because the servers hosting the virtual machines in the cloud mean that the servers are not hosted on the same network as the clients.

Platform as a service (PaaS) is incorrect because PaaS is typically a solution that would deploy hardware and applications platforms for the clients to use.

Infrastructure as a service (IaaS) is incorrect because IaaS is deploying IT hardware through a service provider.

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Question 20

✕ Incorrect

A startup company does not have the capital to invest in infrastructure to host its website, nor does it have the capital to hire engineers to manage its cloud environment. At this point, strict access control is not a concern for the startup, but they do not want to share resources with other organizations. Which cloud implementation will best suit the startup's needs?

☐ Public cloud ✓ Correct☐ Community cloud☐ Hybrid cloud☒ Private cloud ✕ Incorrect**Explanation**

Public cloud will best meet the startup's needs because it offers subscriptions and pay-as-you-go finances and can even provide lower-tier services free of charge. Public clouds are also offered over an internet connection by cloud service providers (CSPs).

Private cloud is unnecessary for the startup's use case because it does not require strict access control in its operations.

Community cloud is incorrect because the startup does not want to share resources with other organizations.

A hybrid cloud is not the best answer because there is potential for resources to be shared among multiple organizations and the requirement to maintain data on-premises, which the startup cannot afford.

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